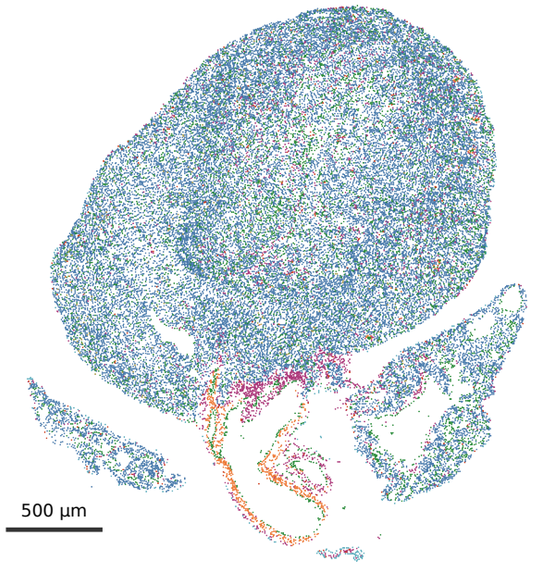
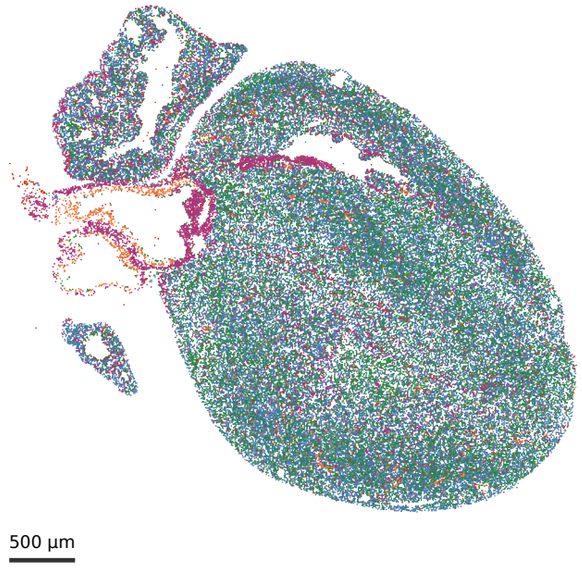
a P0 · broad cardiac identities



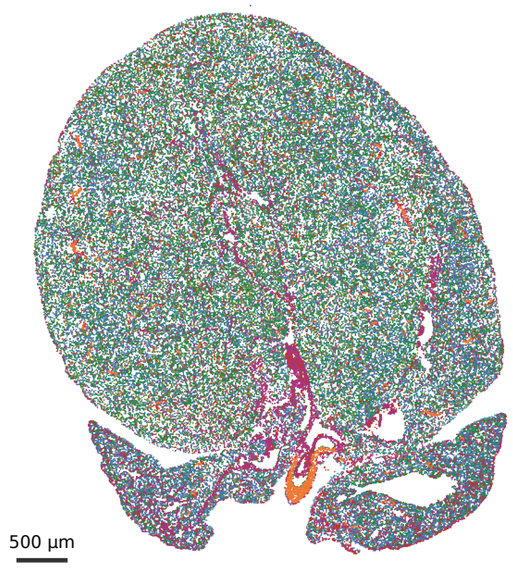
b P7 · broad cardiac identities



c P14 · broad cardiac identities

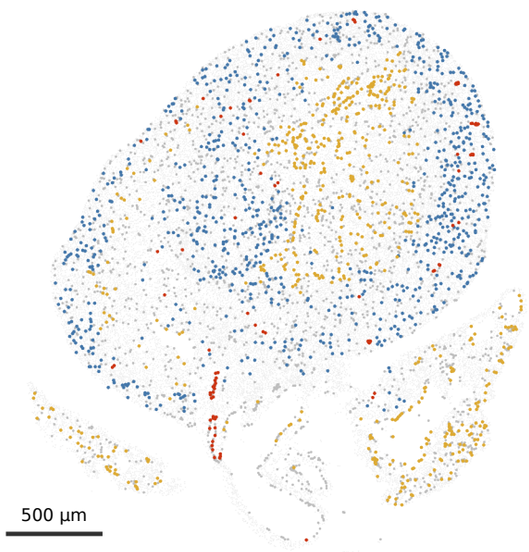


d P21 · broad cardiac identities

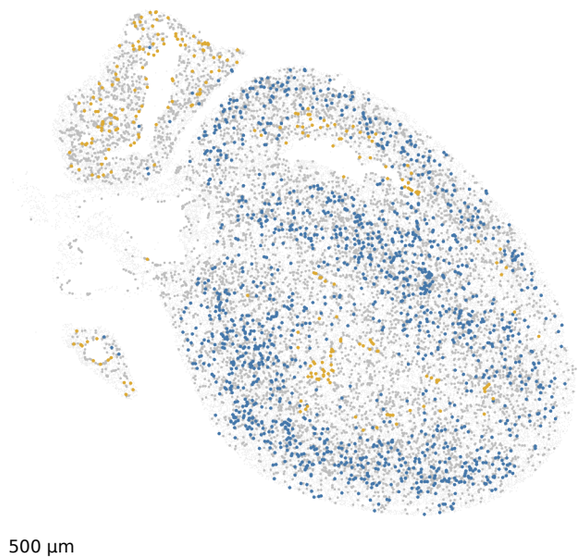


Cardiomyocyte Fibroblast Smooth muscle Epicardial
Endothelial Pericyte Immune

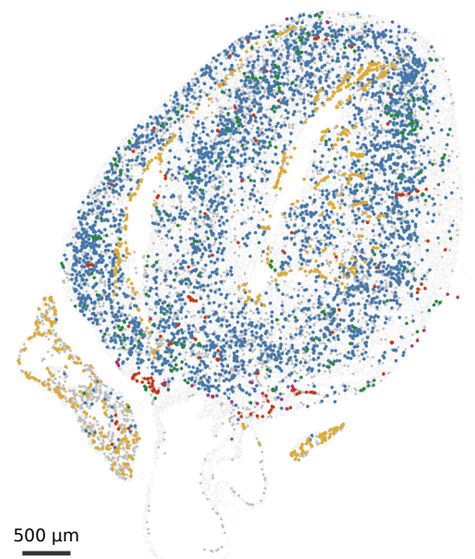
e P0 · supported EC subtypes



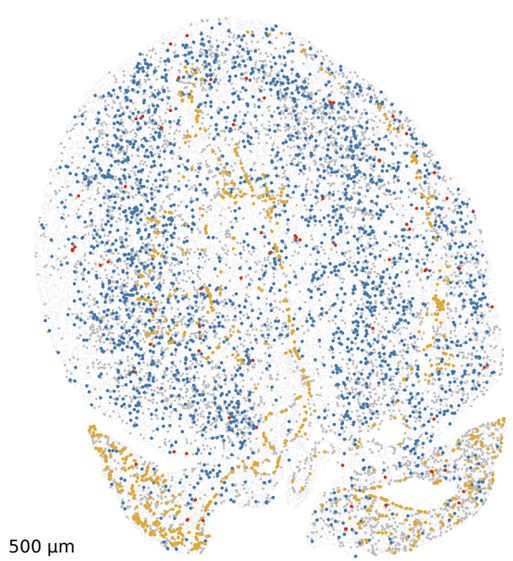
f P7 · supported EC subtypes



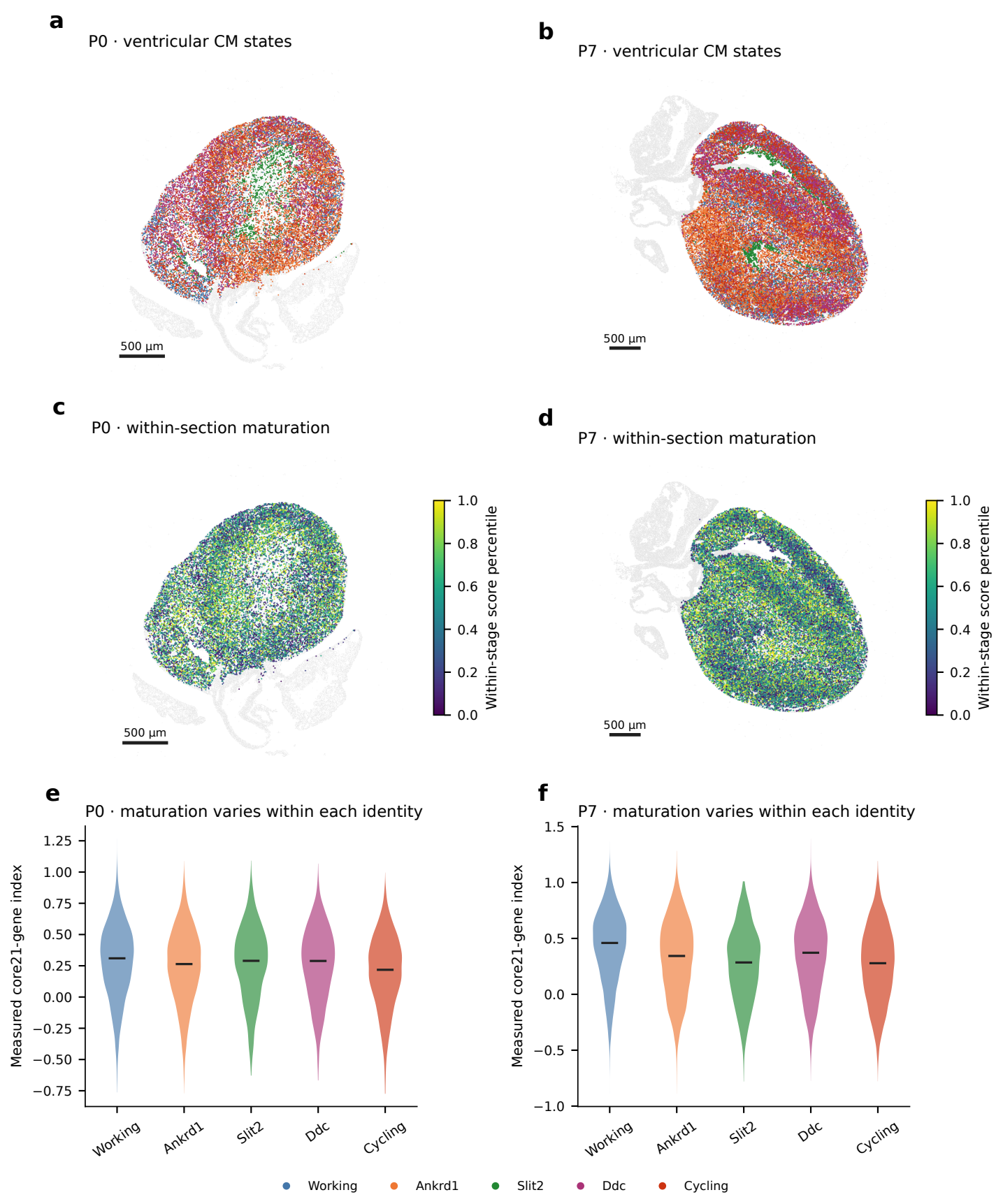
g P14 · supported EC subtypes

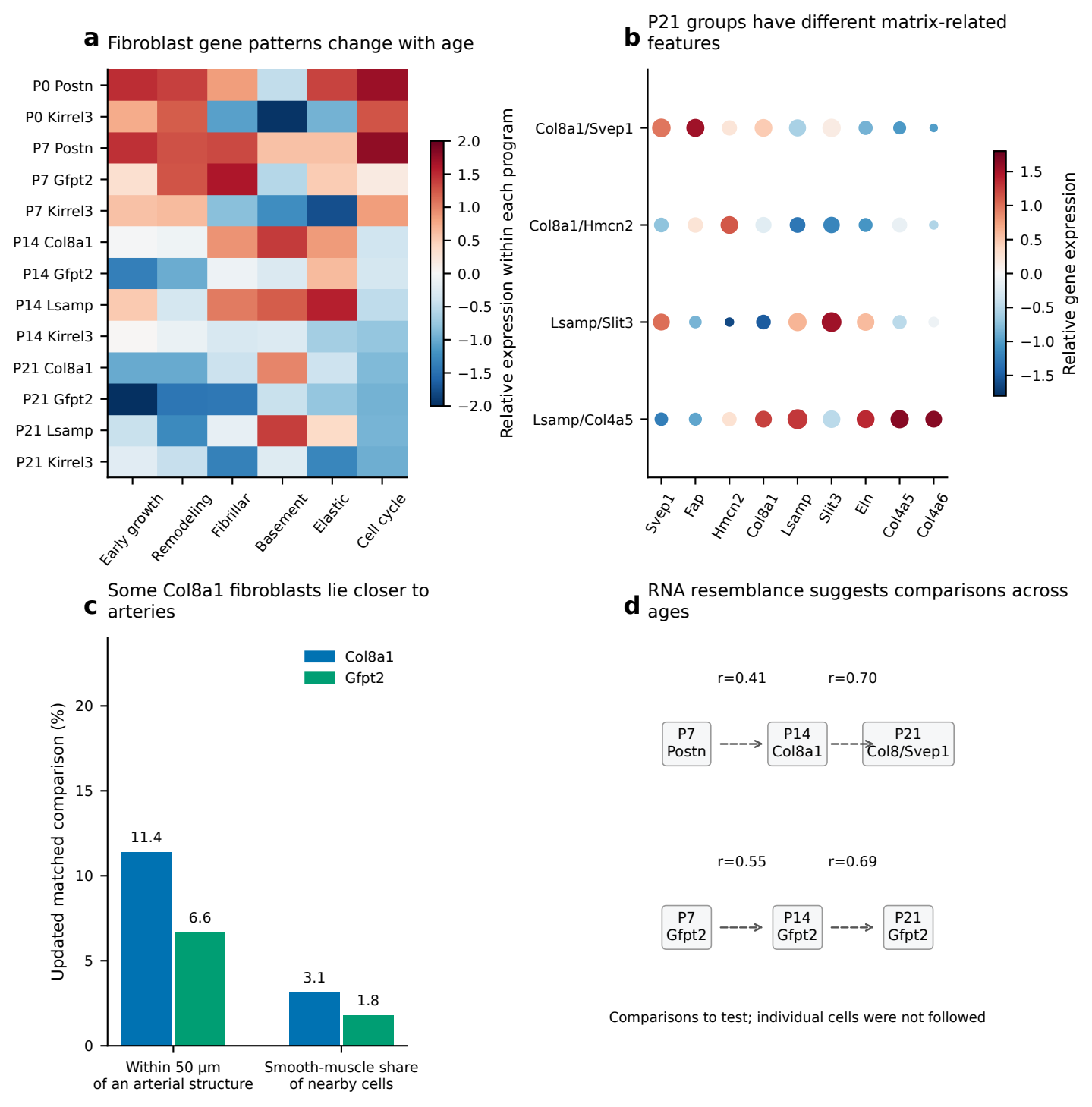


h P21 · supported EC subtypes

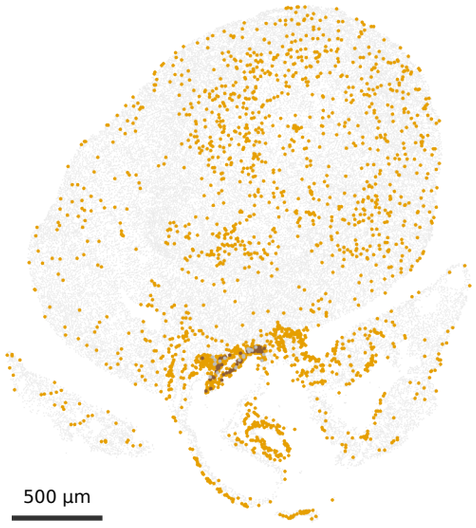


Capillary Endocardial Arterial Venous Lymphatic

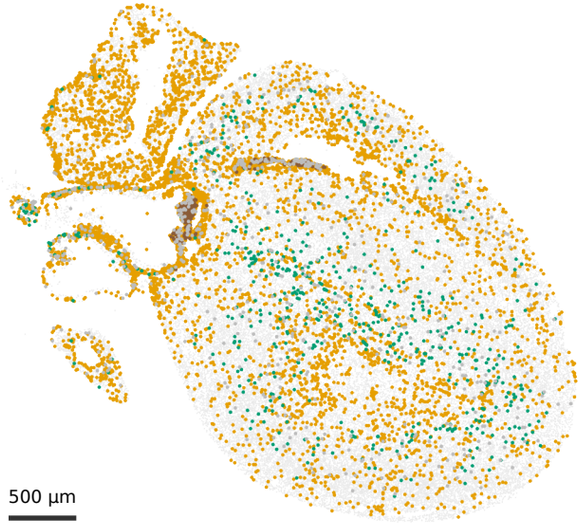

a
P0 · ventricular CM states
500 µm
b
P7 · ventricular CM states
500 µm
c
P0 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
d
P7 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
e
P0 · maturation varies within each identity
Measured core21-gene index
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Working
Ankrd1
Slit2
Ddc
Cycling
f
P7 · maturation varies within each identity
Measured core21-gene index
1.5
1.0
0.5
0.0
−0.5
−1.0
Working
Ankrd1
Slit2
Ddc
Cycling
Working
Ankrd1
Slit2
Ddc
Cycling


a
Fibroblast gene patterns change with age
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early growth
Remodeling
Fibrillar
Basement
Elastic
Cell cycle
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
b
P21 groups have different matrix-related features
Relative expression within each program
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Relative gene expression
c
Some Col8a1 fibroblasts lie closer to arteries
Col8a1
Gfpt2
Updated matched comparison (%)
0
5
10
15
20
11.4
6.6
3.1
1.8
Within 50 µm of an arterial structure
Smooth-muscle share of nearby cells
d
RNA resemblance suggests comparisons across ages
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8/Svep1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Comparisons to test; individual cells were not followed

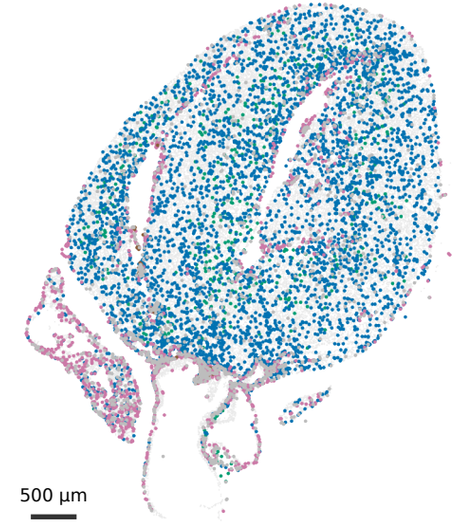
a P0 fibroblast candidates



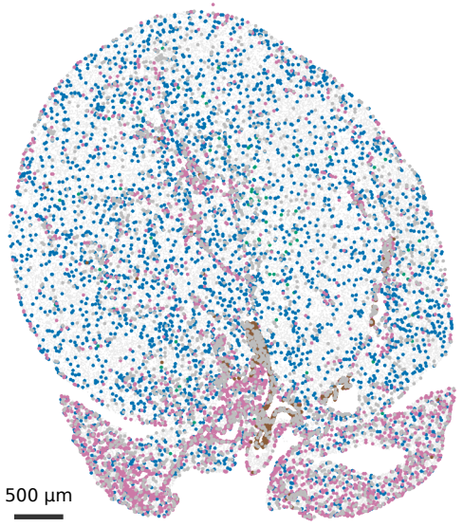
b P7 fibroblast candidates



c P14 fibroblast candidates

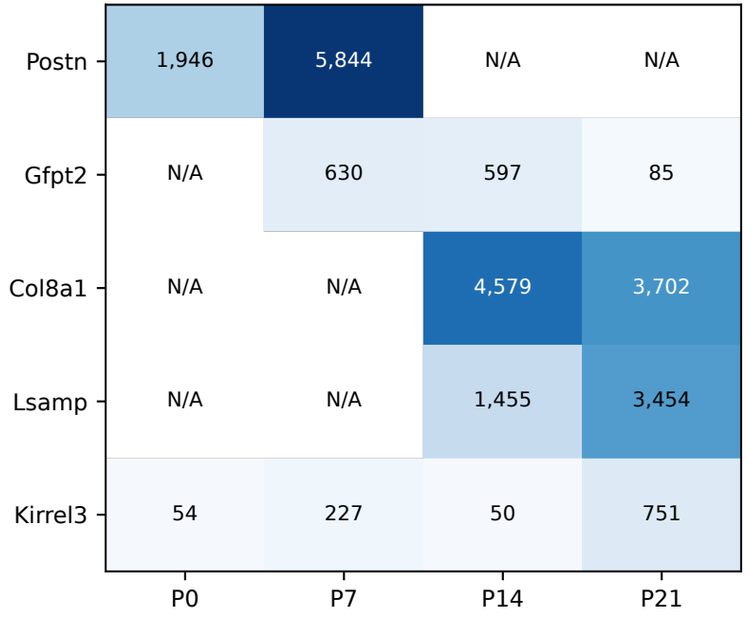


d P21 fibroblast candidates



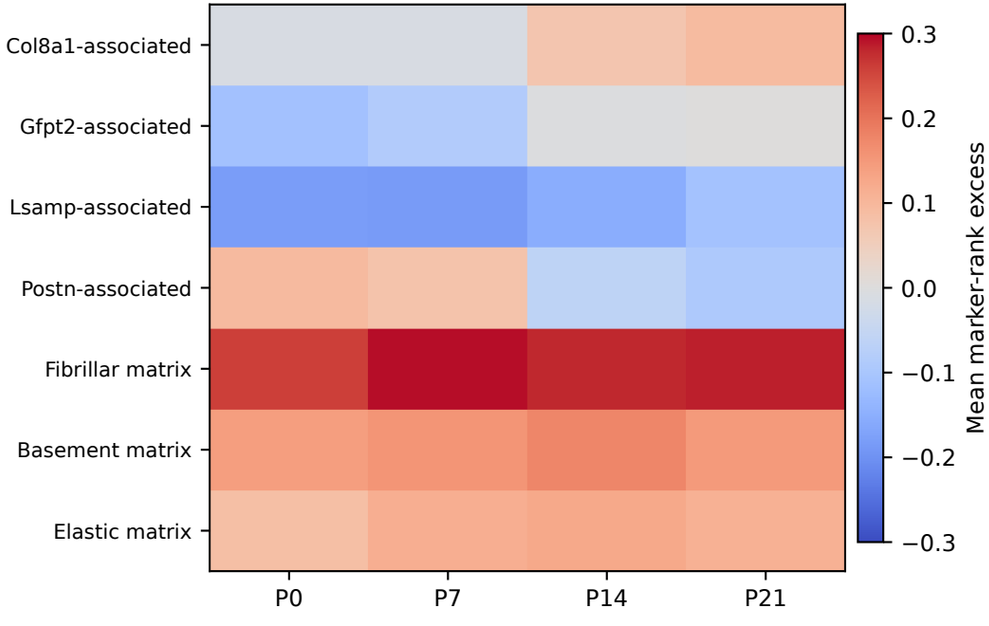
Postn Col8a1 Kirrel3
Gfpt2 Lsamp Unresolved

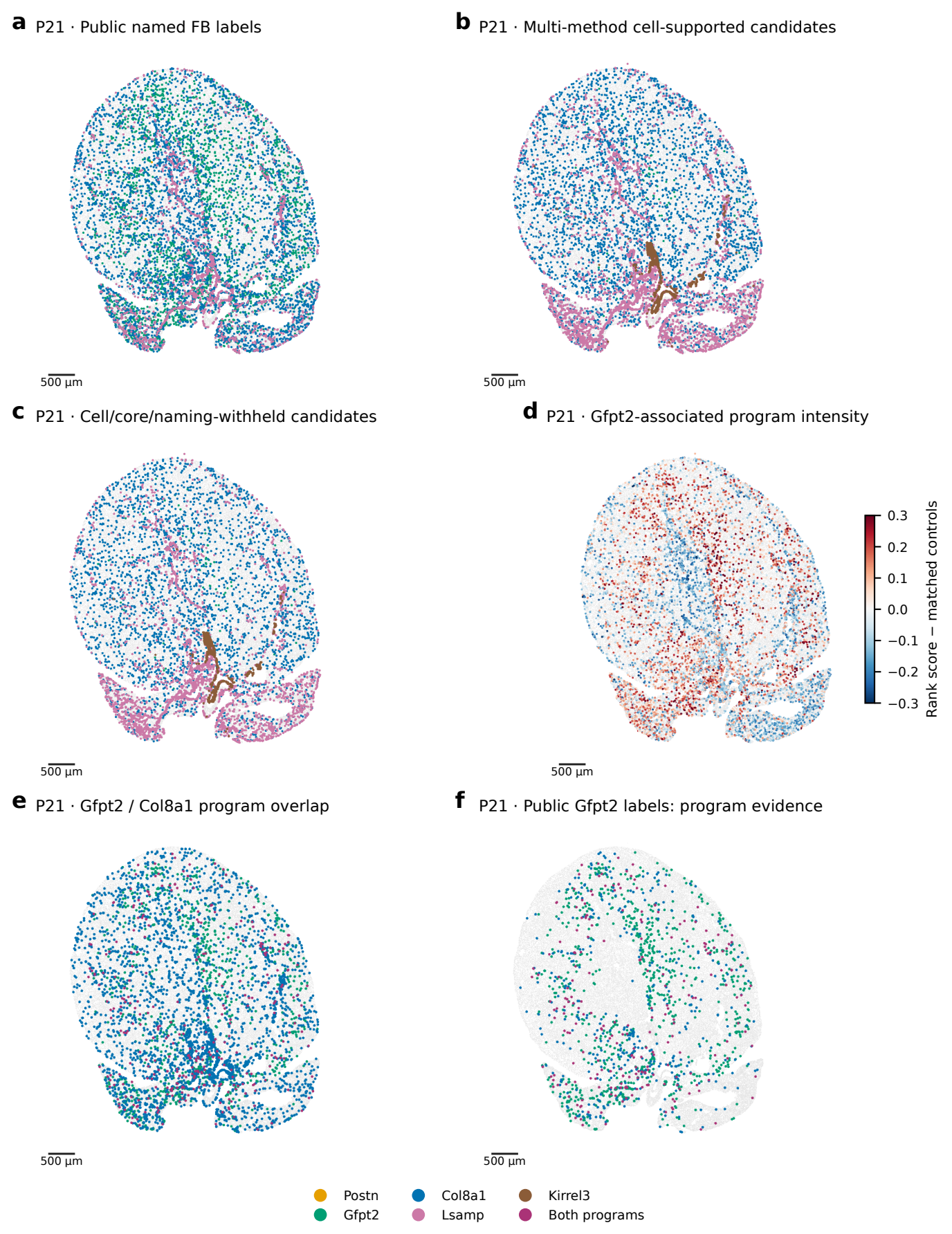
e Assignments require an available reference



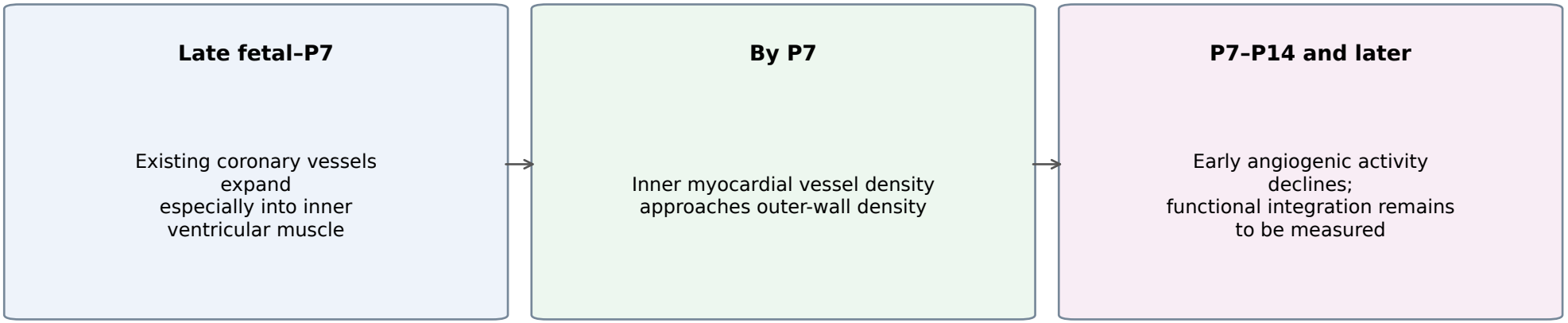
N/A: family not assessable with that age's reference
Counts show coverage, not tissue abundance.

f Gene patterns can be measured at every age



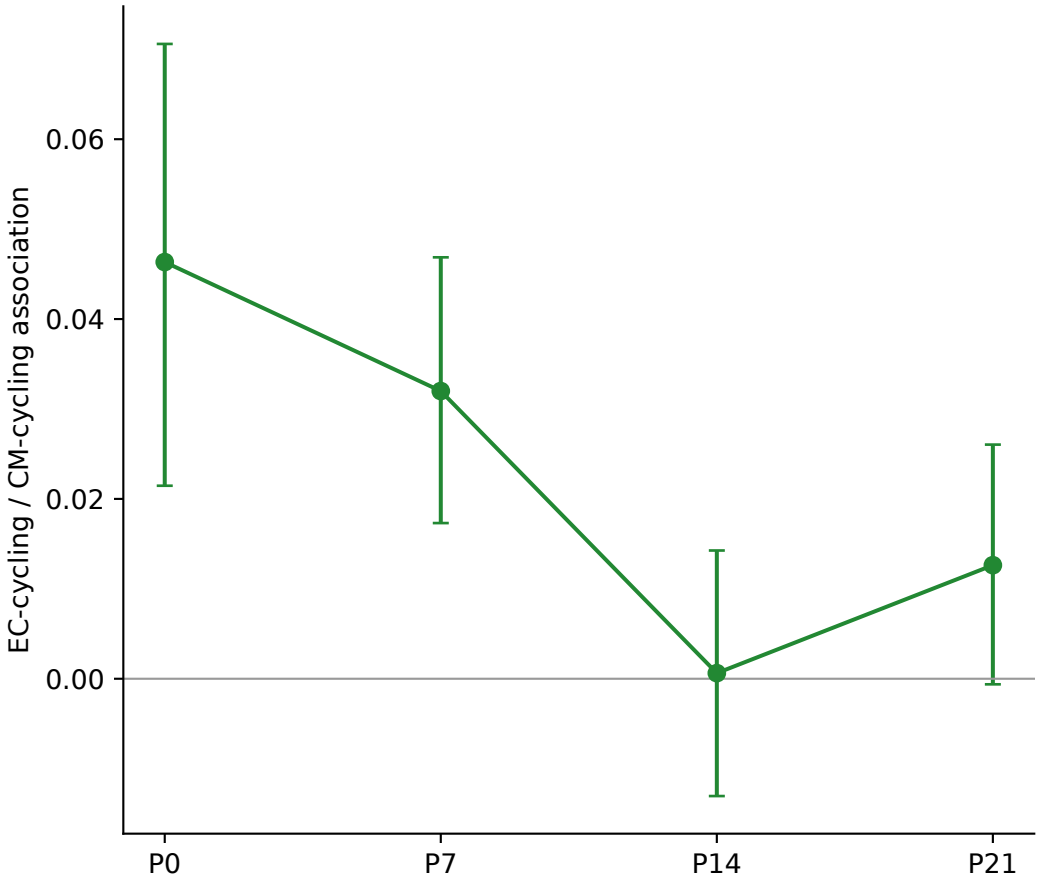

a
P21 · Public named FB labels
500 µm
b
P21 · Multi-method cell-supported candidates
500 µm
c
P21 · Cell/core/naming-withheld candidates
500 µm
d
P21 · Gfpt2-associated program intensity
500 µm
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Rank score − matched controls
e
P21 · Gfpt2 / Col8a1 program overlap
500 µm
f
P21 · Public Gfpt2 labels: program evidence
500 µm
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
Both programs

a Coronary growth provides an anatomical framework

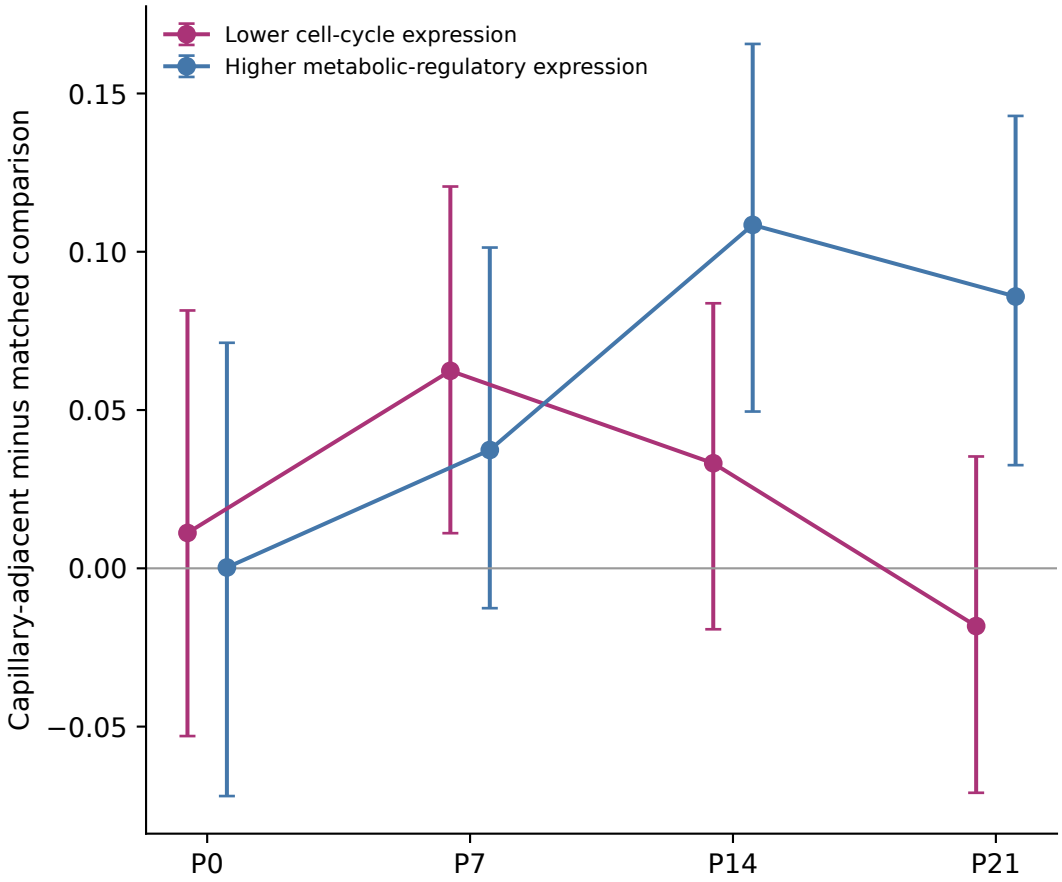


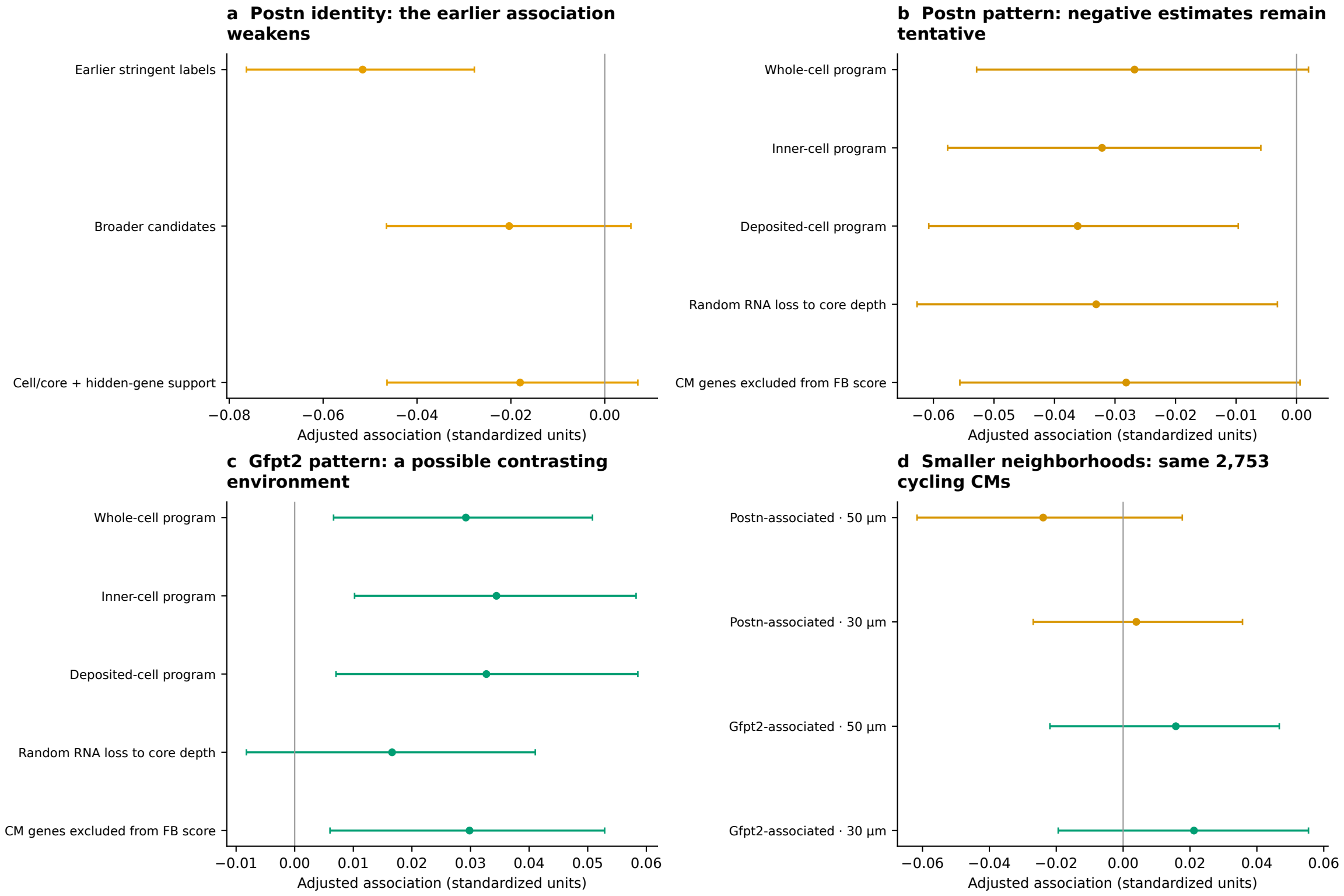
Published vascular sequence from Lu et al. (2021); not new perfusion measurements

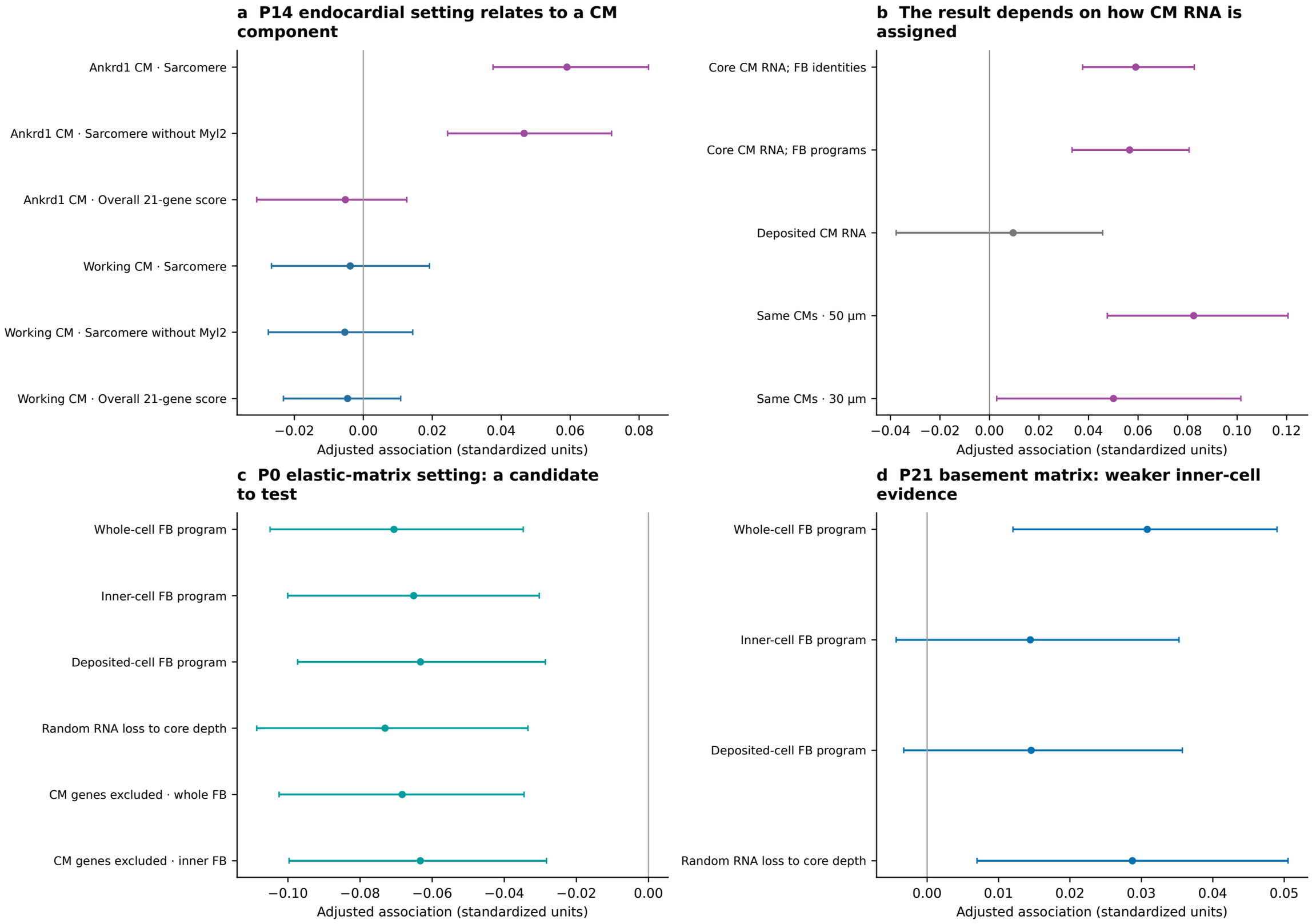
b Local muscle and vascular growth occur together early

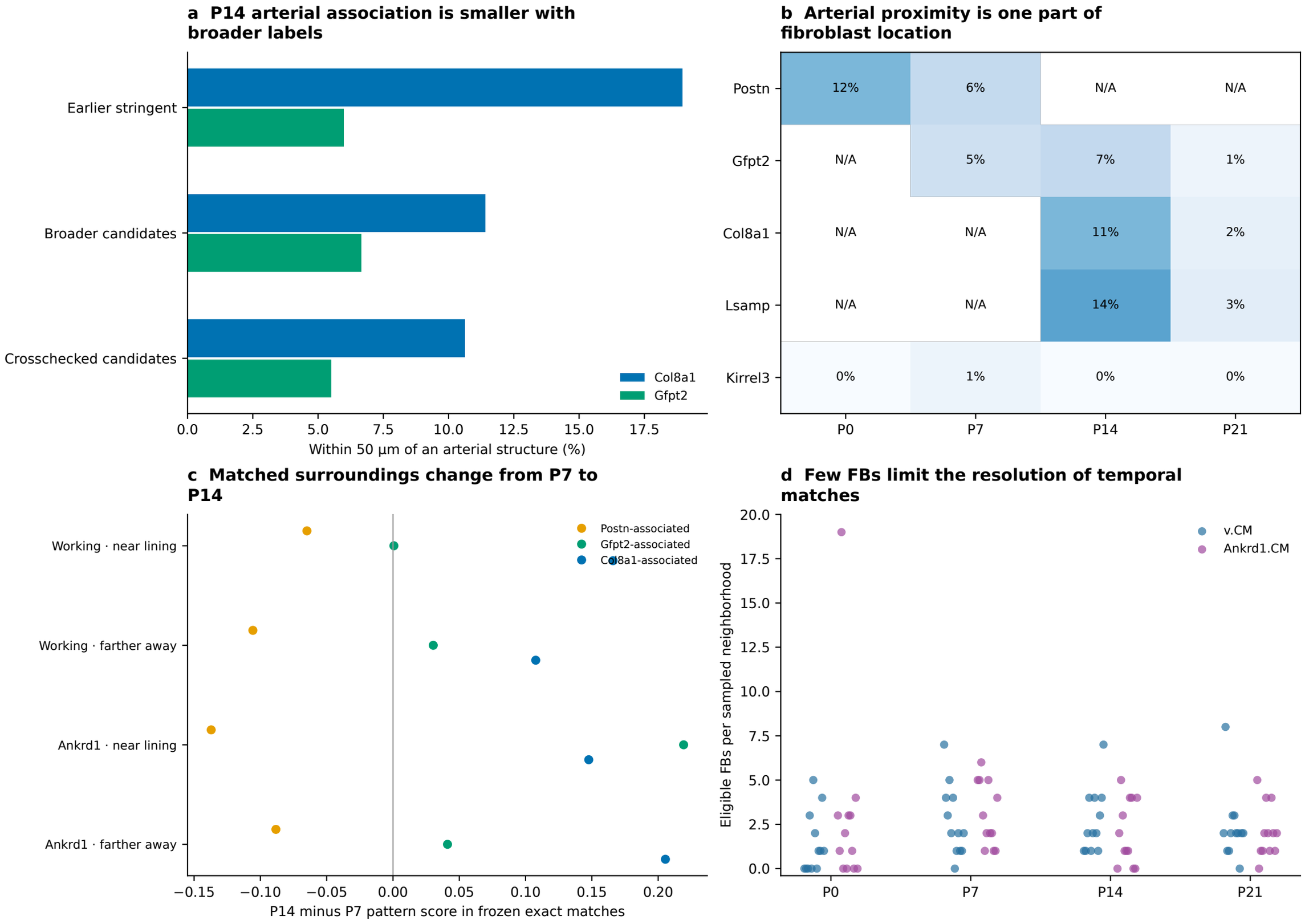


c Capillary proximity relates to different CM components




a Postn identity: the earlier association weakens
Earlier stringent labels
Broader candidates
Cell/core + hidden-gene support
−0.08
−0.06
−0.04
−0.02
0.00
Adjusted association (standardized units)
b Postn pattern: negative estimates remain tentative
Whole-cell program
Inner-cell program
Deposited-cell program
Random RNA loss to core depth
CM genes excluded from FB score
−0.06
−0.05
−0.04
−0.03
−0.02
−0.01
0.00
Adjusted association (standardized units)
c Gfpt2 pattern: a possible contrasting environment
Whole-cell program
Inner-cell program
Deposited-cell program
Random RNA loss to core depth
CM genes excluded from FB score
−0.01
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Adjusted association (standardized units)
d Smaller neighborhoods: same 2,753 cycling CMs
Postn-associated · 50 µm
Postn-associated · 30 µm
Gfpt2-associated · 50 µm
Gfpt2-associated · 30 µm
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
Adjusted association (standardized units)


a P14 endocardial setting relates to a CM component
Ankrd1 CM · Sarcomere
Ankrd1 CM · Sarcomere without Myl2
Ankrd1 CM · Overall 21-gene score
Working CM · Sarcomere
Working CM · Sarcomere without Myl2
Working CM · Overall 21-gene score
−0.02
0.00
0.02
0.04
0.06
0.08
Adjusted association (standardized units)
b The result depends on how CM RNA is assigned
Core CM RNA; FB identities
Core CM RNA; FB programs
Deposited CM RNA
Same CMs · 50 µm
Same CMs · 30 µm
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Adjusted association (standardized units)
c P0 elastic-matrix setting: a candidate to test
Whole-cell FB program
Inner-cell FB program
Deposited-cell FB program
Random RNA loss to core depth
CM genes excluded · whole FB
CM genes excluded · inner FB
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
Adjusted association (standardized units)
d P21 basement matrix: weaker inner-cell evidence
Whole-cell FB program
Inner-cell FB program
Deposited-cell FB program
Random RNA loss to core depth
0.00
0.01
0.02
0.03
0.04
0.05
Adjusted association (standardized units)

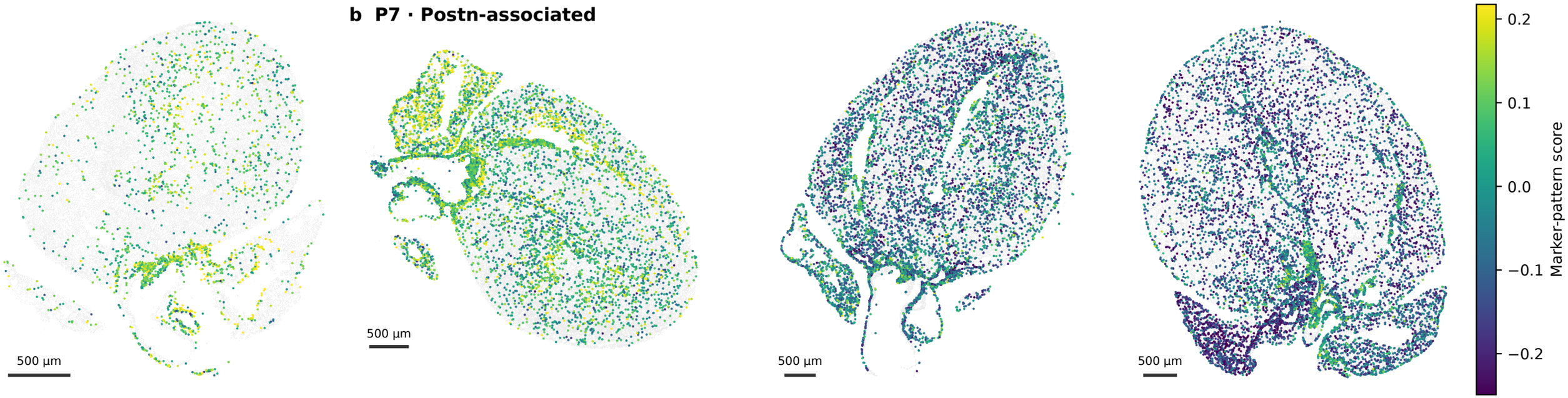

a P14 arterial association is smaller with broader labels
Earlier stringent
Broader candidates
Crosschecked candidates
Col8a1
Gfpt2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Within 50 µm of an arterial structure (%)
b Arterial proximity is one part of fibroblast location
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
12%
6%
N/A
N/A
N/A
5%
7%
1%
N/A
N/A
11%
2%
N/A
N/A
14%
3%
0%
1%
0%
0%
P0
P7
P14
P21
c Matched surroundings change from P7 to P14
Working · near lining
Working · farther away
Ankrd1 · near lining
Ankrd1 · farther away
Postn-associated
Gfpt2-associated
Col8a1-associated
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
P14 minus P7 pattern score in frozen exact matches
d Few FBs limit the resolution of temporal matches
Eligible FBs per sampled neighborhood
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
v.CM
Ankrd1.CM
P0
P7
P14
P21

a P0 · Postn-associated

b P7 · Postn-associated

c P14 · Postn-associated

d P21 · Postn-associated

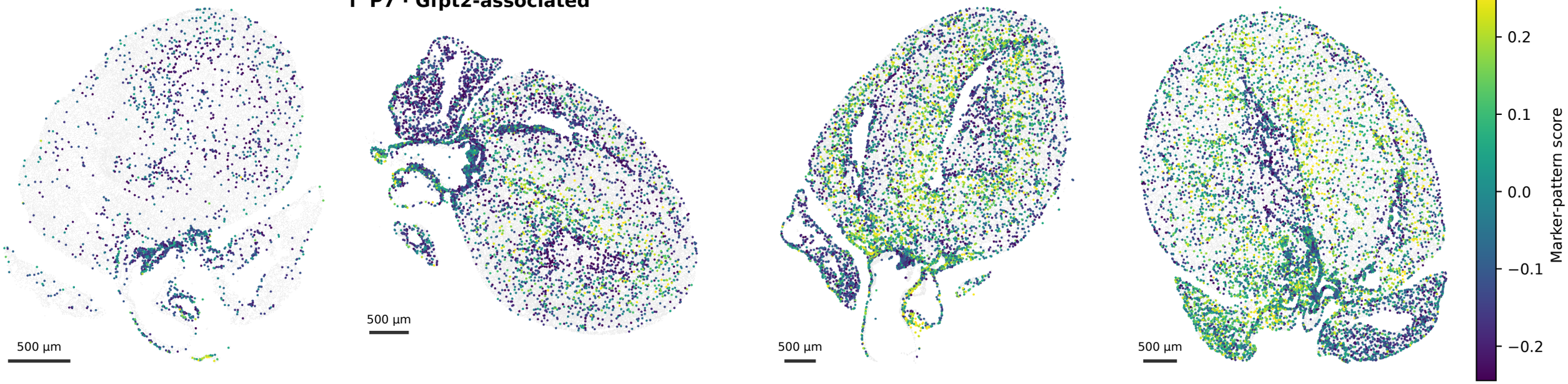


e P0 · Gfpt2-associated

f P7 · Gfpt2-associated

g P14 · Gfpt2-associated

h P21 · Gfpt2-associated

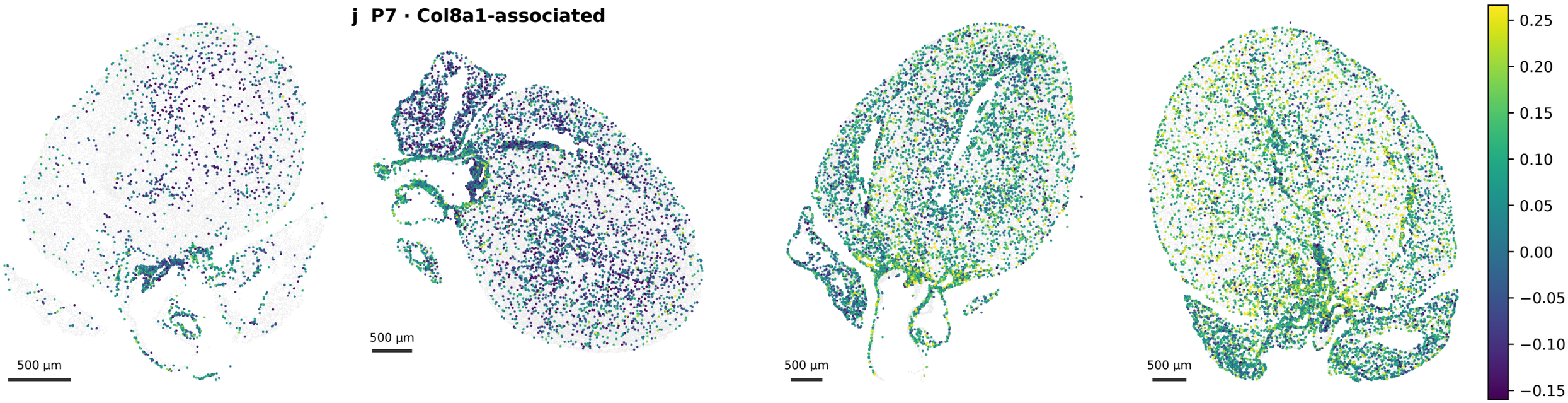


i P0 · Col8a1-associated

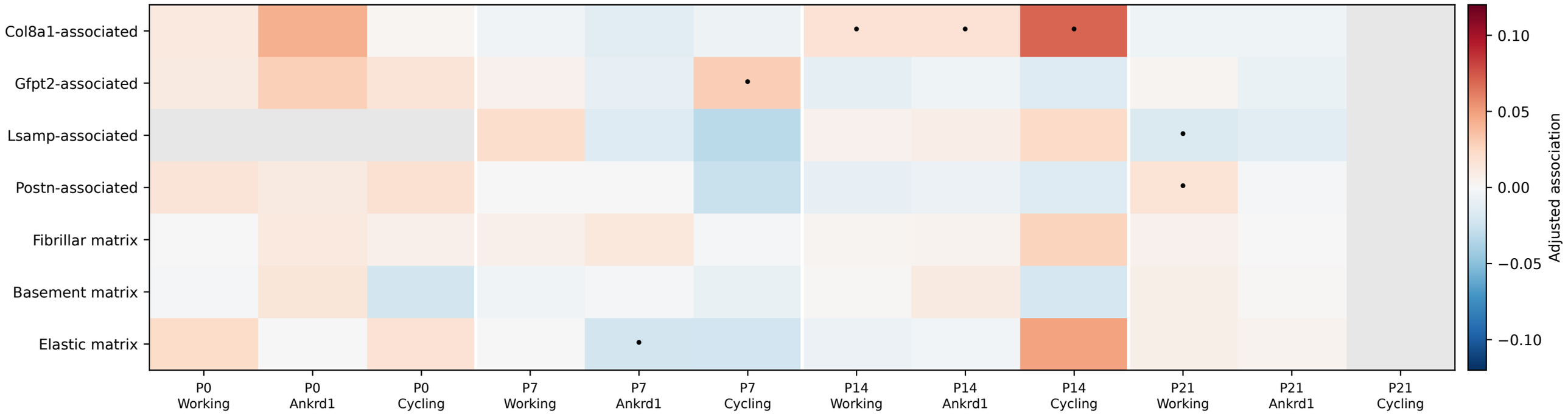
j P7 · Col8a1-associated

k P14 · Col8a1-associated

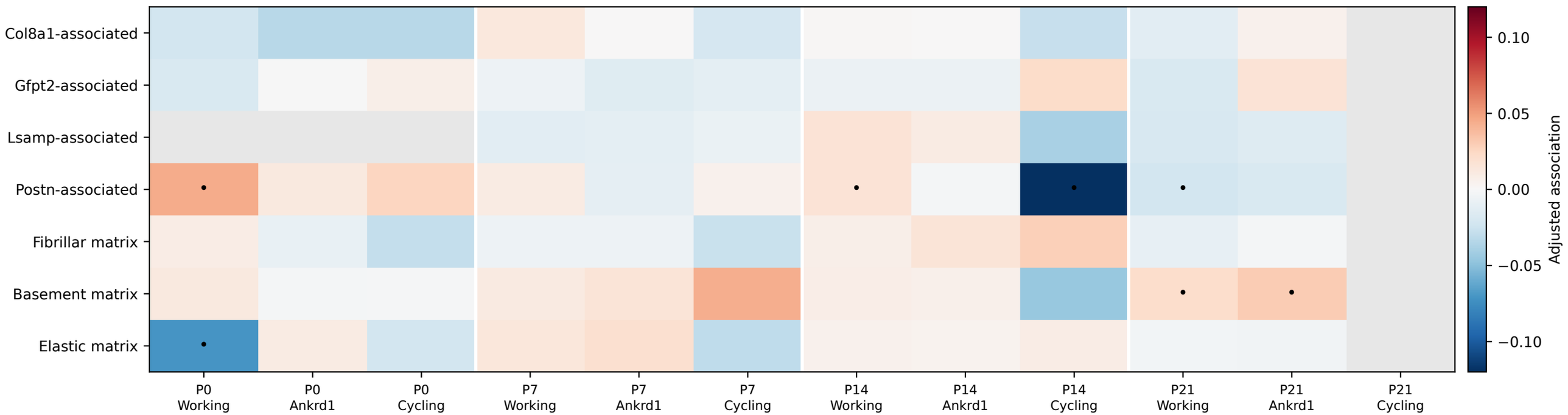
l P21 · Col8a1-associated



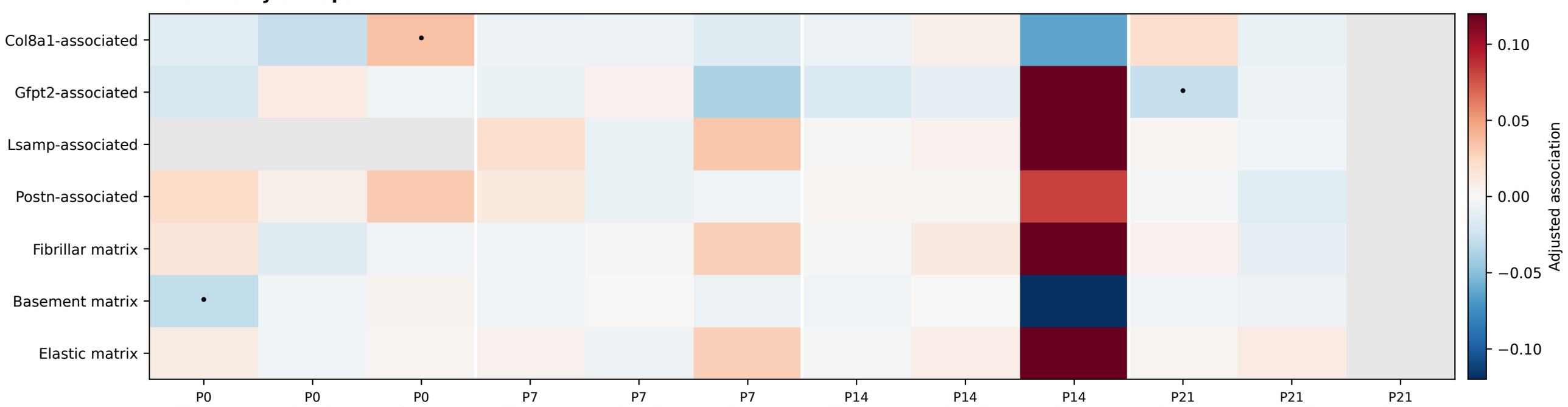
a CM energy-metabolism expression

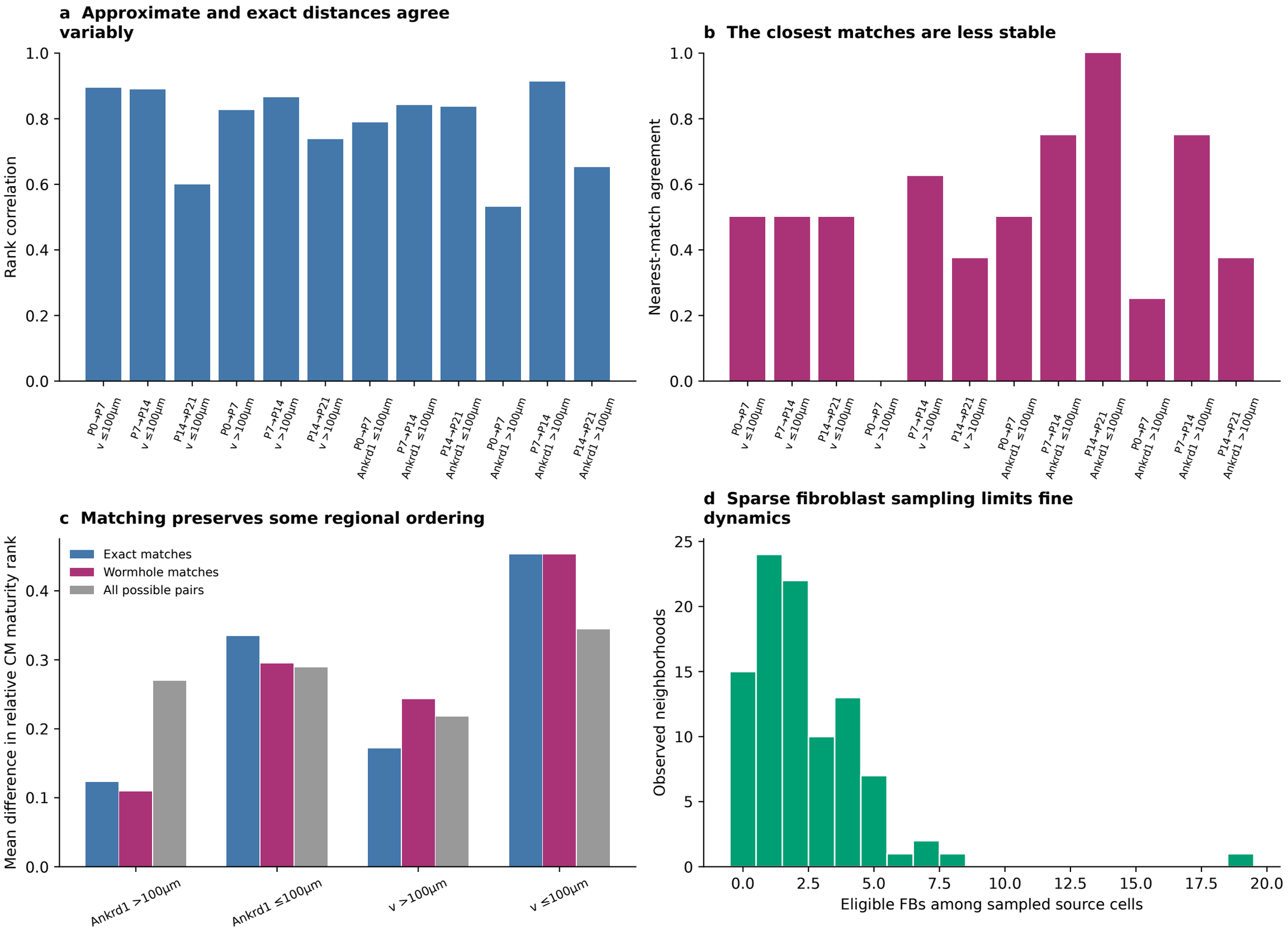


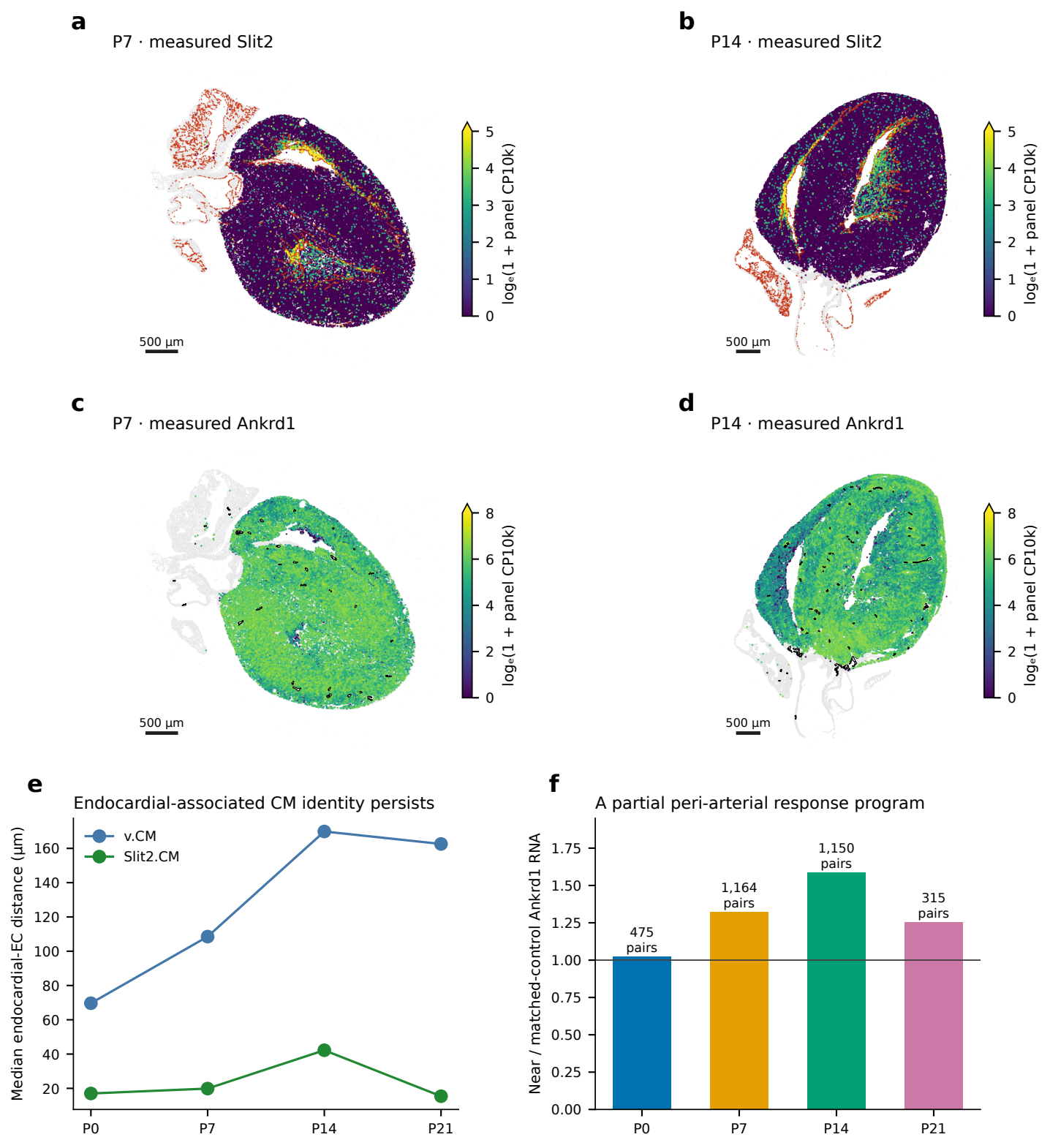
b CM sarcomere expression



c CM cell-cycle expression




a Approximate and exact distances agree variably
Rank correlation
1.0
0.8
0.6
0.4
0.2
0.0
P0→P7 v ≤100μm
P7→P14 v ≤100μm
P14→P21 v ≤100μm
P0→P7 v >100μm
P7→P14 v >100μm
P14→P21 v >100μm
P0→P7 Ankrd1 ≤100μm
P7→P14 Ankrd1 ≤100μm
P14→P21 Ankrd1 ≤100μm
P0→P7 Ankrd1 >100μm
P7→P14 Ankrd1 >100μm
P14→P21 Ankrd1 >100μm
b The closest matches are less stable
Nearest-match agreement
1.0
0.8
0.6
0.4
0.2
0.0
P0→P7 v ≤100μm
P7→P14 v ≤100μm
P14→P21 v ≤100μm
P0→P7 v >100μm
P7→P14 v >100μm
P14→P21 v >100μm
P0→P7 Ankrd1 ≤100μm
P7→P14 Ankrd1 ≤100μm
P14→P21 Ankrd1 ≤100μm
P0→P7 Ankrd1 >100μm
P7→P14 Ankrd1 >100μm
P14→P21 Ankrd1 >100μm
c Matching preserves some regional ordering
Exact matches
Wormhole matches
All possible pairs
Mean difference in relative CM maturity rank
0.4
0.3
0.2
0.1
0.0
Ankrd1 >100μm
Ankrd1 ≤100μm
v >100μm
v ≤100μm
d Sparse fibroblast sampling limits fine dynamics
Observed neighborhoods
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Eligible FBs among sampled source cells


a
P7 · measured Slit2
500 µm
log$_e$(1 + panel CP10k)
b
P14 · measured Slit2
500 µm
log$_e$(1 + panel CP10k)
c
P7 · measured Ankrd1
500 µm
log$_e$(1 + panel CP10k)
d
P14 · measured Ankrd1
500 µm
log$_e$(1 + panel CP10k)
e
Endocardial-associated CM identity persists
v.CM
Slit2.CM
Median endocardial-EC distance (µm)
P0
P7
P14
P21
f
A partial peri-arterial response program
Near / matched-control Ankrd1 RNA
475 pairs
1,164 pairs
1,150 pairs
315 pairs
P0
P7
P14
P21